

SOUTHERN GAS CORPORATION

Cuddalore Gas Processing Plant — Compression Train 3

RECIPROCATING COMPRESSOR — HEALTH & PREDICTIVE MAINTENANCE REPORT

Report No.	SGC-PM-2026-00345	Date	05-Sep-2026
Equipment Tag	K-301A	Train	Compression Train 3
Equipment Type	Reciprocating Compressor (2-Stage)	Work Order	WO-2026-301188
Manufacturer	Dresser-Rand	Model	HHE-VG / 2-Stage
Installed On	03-Jun-2014	Running Hours	68,400 hrs
Inspector	T. Krishnamurthy (Lead Mech. Tech)	Emp ID	EMP-20891
Process Engineer	Dr. A. Balasubramanian	Approval Date	06-Sep-2026

1. Equipment Description & Process Context

Compressor K-301A is a two-stage reciprocating compressor serving the gas sweetening section of the Cuddalore Gas Processing Plant. It compresses treated natural gas from 12 bar suction to 72 bar discharge at a design throughput of 180,000 Nm3/day. The machine is driven by an Siemens 1LG6-357 induction motor (2,200 kW, 11 kV) through a direct coupling. Piston rings on Stage 1 cylinders were last replaced during the 2024 annual turnaround (TA-2024). Stage 2 suction valve assemblies showed early leakage signatures in the Q2-2026 thermal imaging survey (TI-SGC-2026-Q2-018).

2. Predictive Maintenance Sensor Data

Sensor / Parameter	Tag	Current Value	Alarm Limit	Status
Stage 1 Suction Pressure	PT-K301A-S1S	12.4 bar	11.5 – 13.0 bar	Normal
Stage 1 Discharge Pressure	PT-K301A-S1D	32.1 bar	30 – 34 bar	Normal
Stage 2 Suction Pressure	PT-K301A-S2S	31.6 bar	30 – 34 bar	Normal
Stage 2 Discharge Pressure	PT-K301A-S2D	70.4 bar	68 – 74 bar	Normal
Interstage Gas Temperature	TT-K301A-INT	98 °C	≤ 105 °C	Normal
Final Discharge Temperature	TT-K301A-DIS	112 °C	≤ 115 °C	Marginal
Crankshaft Vibration X-axis	VT-K301A-CRK-X	4.2 mm/s	≤ 4.0 mm/s	Above Alarm
Crankshaft Vibration Y-axis	VT-K301A-CRK-Y	2.8 mm/s	≤ 4.0 mm/s	Normal
Stage 2 Valve Temperature (IR)	TT-K301A-VLV-S2	127 °C	≤ 120 °C	Above Alarm
Lube Oil Pressure	PT-K301A-OIL	2.8 bar	2.5 – 3.5 bar	Normal
Lube Oil Temperature	TT-K301A-OIL	61 °C	≤ 70 °C	Normal
Motor Winding Temperature	TT-MOT-K301A	88 °C	≤ 95 °C	Normal
Rod Drop — Cylinder 1	RD-K301A-C1	0.38 mm	≤ 0.50 mm	Normal

Rod Drop — Cylinder 3	RD-K301A-C3	0.48 mm	≤ 0.50 mm	Marginal
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3. Issues Identified

Three issues require attention based on current readings and trend analysis:

Issue 1 — Stage 2 Suction Valve Leakage (K301A-VLV-S2): Thermal imaging and high valve temperature (127 deg C vs. 120 deg C alarm) confirm Stage 2 suction valve assembly is leaking. This is consistent with the Q2-2026 survey findings. Leaking valves reduce volumetric efficiency and increase discharge temperature. Valve replacement is recommended at next planned maintenance window.

Issue 2 — Crankshaft Vibration (X-axis above alarm): Vibration of 4.2 mm/s on the crankshaft X-axis exceeds the 4.0 mm/s alarm level. Root cause is likely related to connecting rod bearing wear on cylinder 1 given the marginal rod drop reading (0.48 mm). A spectrographic oil analysis was collected and sent to lab (LAB-OIL-2026-0844).

Issue 3 — Cylinder 3 Rod Drop Trending (Marginal): Rod drop on cylinder 3 is 0.48 mm against a limit of 0.50 mm. This indicates piston rider band wear. If not addressed by the next turnaround, unplanned shutdown risk increases significantly.

4. Materials & Lubrication

Cylinder lubrication uses Petro-Canada Purity FG Compressor Oil ISO VG 100 (food-grade). Crankcase uses Shell Tellus S4 ME 46 hydraulic oil (ISO VG 46). Piston rings (Stage 1) are PTFE-filled carbon grade from Hoerbiger. Stage 2 suction and discharge valves are Hoerbiger RINO plate valves.

5. Technician Field Notes

"Compressor was inspected during a 2-hour planned window. Audible higher-frequency tone near Stage 2 head confirms the valve condition we saw in the thermal survey. Rod drop on C3 is very close to the limit — I would not delay the rider band inspection beyond this TA cycle. The crankshaft vibration spike is new since last month; watching it closely. Lube oil looks clean, colour normal, no metallic sheen visible in the drain sample — lab results will confirm. Motor and coupling are fine." — T. Krishnamurthy, Lead Mech. Tech

6. Recommended Actions

#	Action	Priority	Target Date	Owner
1	Replace Stage 2 suction valve assembly (K301A-VLV-S2)	High	Next planned window	T. Krishnamurthy
2	Review lube oil spectrographic results (LAB-OIL-2026-0844)	Medium	12-Sep-2026	Reliability Team
3	Inspect connecting rod bearing — Cylinder 1 during TA	High	TA-2026 (Oct)	Lead Technician
4	Schedule rider band replacement — Cylinder 3 in High	High	TA-2026 (Oct)	T. Krishnamurthy
5	Weekly vibration trending on crankshaft until TA	Medium	Ongoing	Condition Monitoring